Gidea–Katz (2018): methodological pipeline

4 US equity indices
S&P 500 · DJIA · NASDAQ · Russell 2000

→ Daily log returns

→ Sliding window

→ Point cloud in R^4

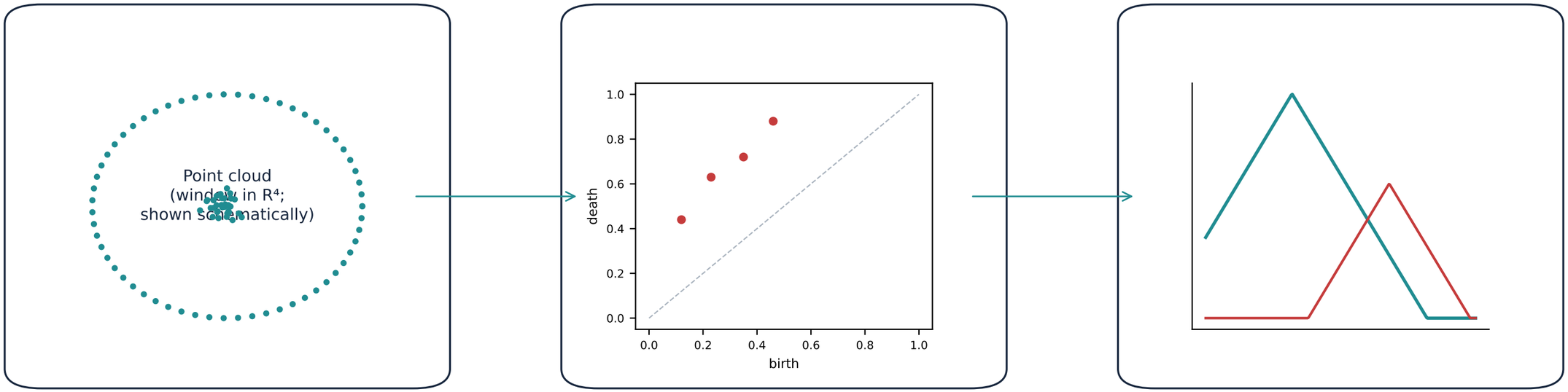
→ Vietoris–Rips filtration

→ Persistent H1

→ Persistence diagram

→ Persistence landscape

→ L^p norm

→ Early-warning signal

Conceptual transformation (not an original-paper figure)